

Problem-Solution Fit

Field	Details
Date	05-07-2026
Team ID	(Your Team ID)
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas

Section	Content
1. CUSTOMER SEGMENT(S) [CS]	Farmers, Agricultural Students, Farm Consultants, Government Agriculture Departments
2. PROBLEMS / PAINS + ITS FREQUENCY [PR]	Farmers frequently struggle to select the most suitable crop based on soil nutrients, pH, temperature, humidity, and rainfall, resulting in reduced crop yield and financial losses.
3. TRIGGERS TO ACT [TR]	Low crop productivity, changing climate conditions, soil fertility issues, need for better farming decisions, increasing demand for smart agriculture.
4. EMOTIONS BEFORE / AFTER [EM]	Before: Confused, uncertain, worried about crop failure. After: Confident, satisfied, optimistic about higher yield and better income.
5. AVAILABLE SOLUTIONS (PROS & CONS) [AS]	Traditional farming advice, agricultural experts, online resources. Cons: Time-consuming, Pros: costly, less personalized, not always accurate.
6. CUSTOMER LIMITATIONS [CL]	Limited technical knowledge, lack of real-time guidance, budget constraints, limited internet access in rural areas.
7. BEHAVIOR + ITS INTENSITY [BE]	Farmers generally rely on traditional experience and local advice. Adoption of digital tools is increasing but still moderate.
8. CHANNELS OF BEHAVIOR [CH]	Online: Web application, Mobile phones, Agricultural websites. Offline: Agriculture officers, Local markets, Farmer meetings, Training programs.

Section	Content
9. PROBLEM ROOT / CAUSE [RC]	Lack of accurate crop recommendation based on scientific soil and environmental analysis, changing weather conditions, insufficient access to expert guidance.
10. YOUR SOLUTION [SL]	OptiCrop is a Machine Learning based web application that recommends the most suitable crop using soil nutrients (N, P, K), temperature, humidity, pH, and rainfall, helping farmers improve productivity and make informed farming decisions.